LEARN AI AGENTS



HRUTHIK GALI DILEEP

FOLLOW

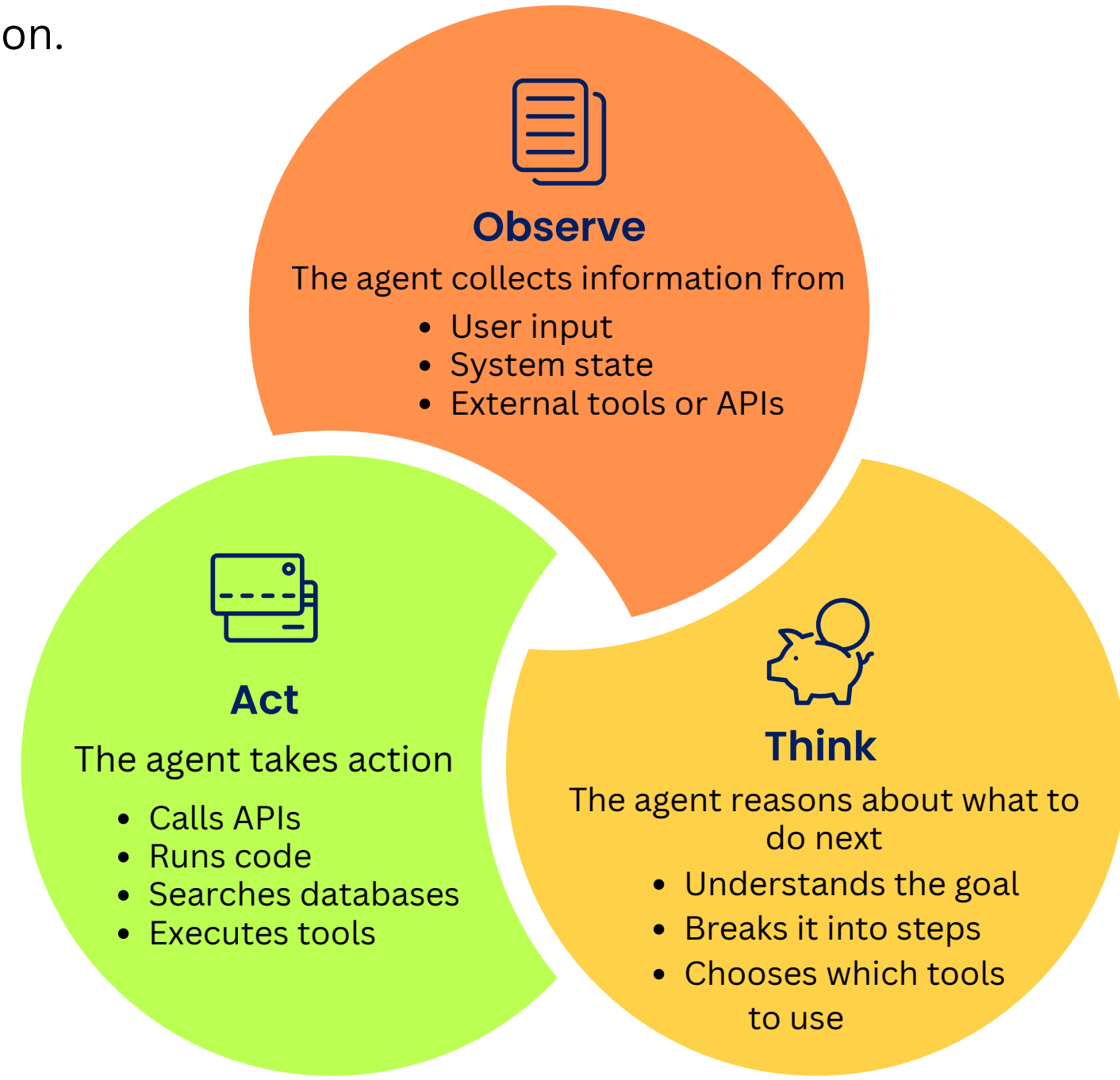
Repost

AI Agents

What are AI Agents?

AI Agent = LLM + Memory + Tools + Control Loop

An AI Agent is an autonomous system that perceives its environment, makes decisions, and takes actions to achieve specific goals without constant human intervention.



Key Characteristics

1. **Autonomy - operates independently**
2. **Perception - gathers environmental information**
3. **Action - executes tasks and decisions**
4. **Goal-oriented - works toward defined objectives**
5. **Adaptability - learns and adjusts behavior**

Types of AI Agents

01

Reactive Agents

Respond to current perceptions without memory.

Example: A thermostat that turns on heating when temperature drops below threshold.

02

Goal-Based Agents

Plan actions to achieve specific goals.

Example: GPS navigation finding the fastest route to your destination.

03

Utility-Based Agents

Maximize utility function, handling trade-offs.

Example: Trading bot balancing risk vs. reward to optimize portfolio returns.

04

Learning Agents

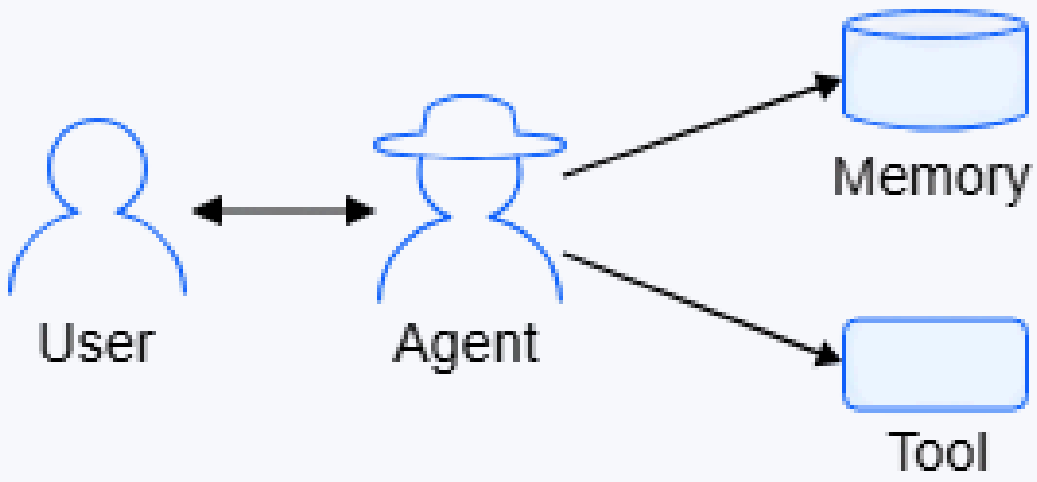
Improve performance over time through experience and feedback.

Example: Recommendation system that gets better at suggesting content based on your viewing history.

Agent Architectures

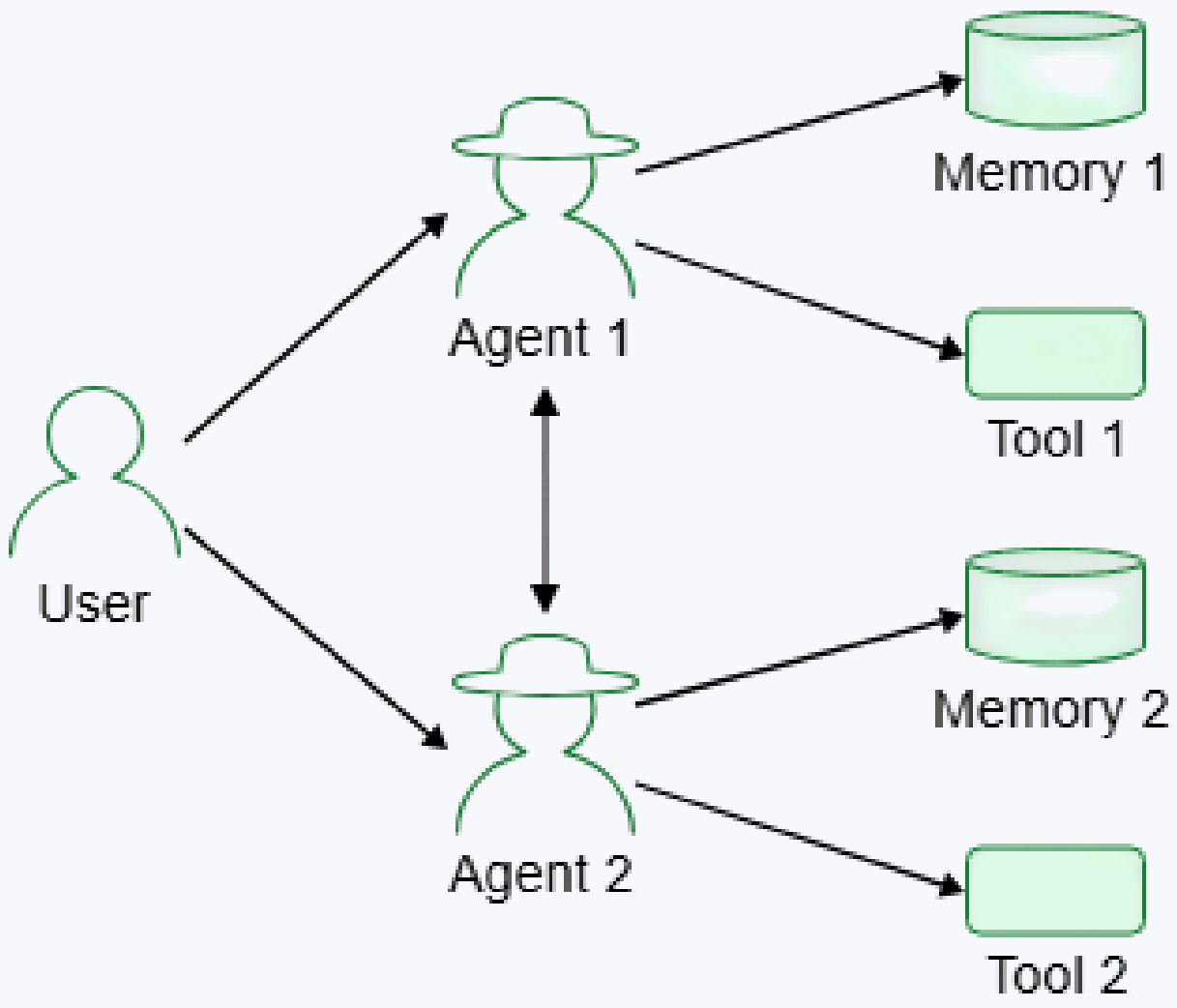
Single Agent Systems

- One autonomous agent operating independently
- Centralized decision-making, simpler coordination
- Example: Virtual assistant like Siri handling voice commands and tasks for a single user



Multi-Agent Systems

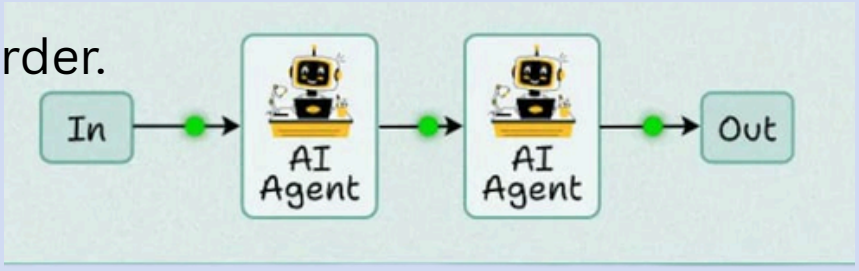
- Multiple agents working cooperatively or competitively
- Distributed intelligence with communication protocols
- Example: Warehouse robots coordinating to optimize package sorting and delivery routes



AI Agent Orchestration Designs

Sequential

Agents execute tasks one after another in order.
Example: Document processing where one agent extracts text, next translates it, then another summarizes.



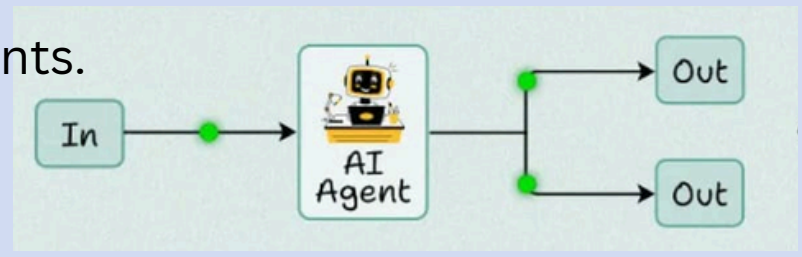
Parallel



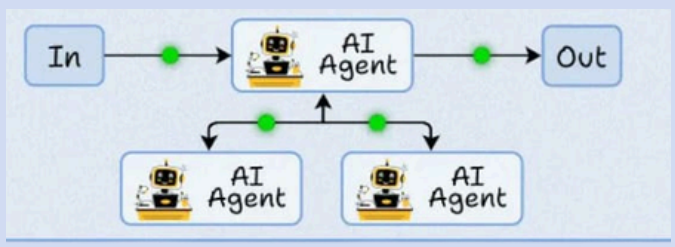
Multiple agents execute tasks simultaneously.
Example: Web scraping where different agents simultaneously collect data from multiple websites.

Router

Central router directs tasks to specialized agents.
Example: Customer service system routing technical queries to tech support agent and billing questions to finance agent.



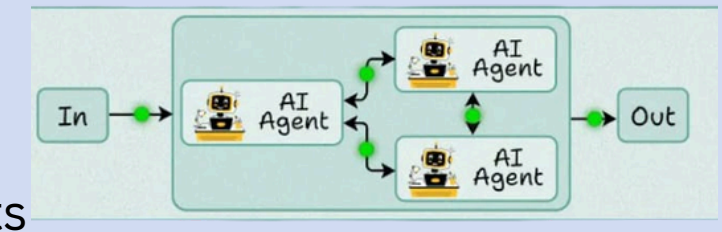
Hierarchical



Manager agents coordinate worker agents in tree structure.
Example: Project management system where manager agent delegates subtasks to specialist worker agents.

Network

Agents in mesh topology with peer-to-peer communication.
Example: Distributed sensor network where agents share information with neighbors to detect patterns.



Agent Design Patterns

ReAct (Reasoning + Acting)

- Interleaves reasoning and action steps
- Transparent thought process with tool usage
- Example: Research assistant that thinks "I need weather data," searches for it, observes results, then formulates answer



Plan and Execute

- Creates complete plan first, then executes systematically
- Best for complex multi-step tasks
- Example: Travel booking agent that plans entire trip (flights, hotels, activities) before making any reservations

Human-in-the-Loop

Humans review or approve agent actions when needed.
Why

- Reduces errors
- Improves reliability
- Keeps control with humans

Example:
AI drafts content → Human reviews → Final output

Reflection Pattern

- Agent reviews and critiques its own outputs
- Iterative refinement for quality improvement
- Example: Code generator that writes code, reviews it for bugs, then refines until tests pass

Tool Calling Agent

- Accesses external tools and APIs dynamically
- Extensible capabilities with grounded results
- Example: Data analyst agent that can query databases, run calculations, and generate visualizations as needed

Memory Management in Agents

Short-Term Memory

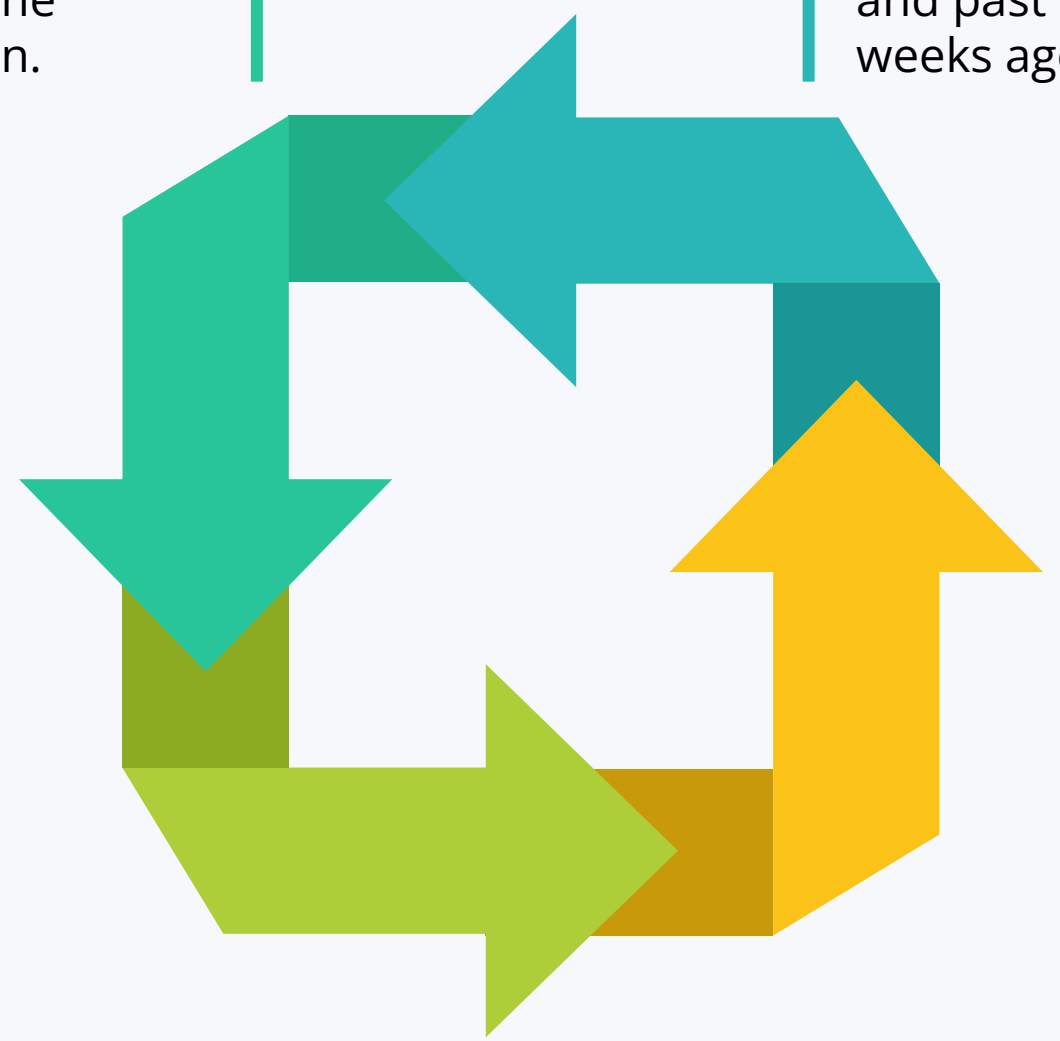
Current conversation context within token limits.

Example: Chatbot remembering what you said 3 messages ago in the current conversation.

Long-Term Memory

Persistent storage across sessions in databases.

Example: Assistant remembering your preferences and past conversations from weeks ago.



Semantic Memory

General knowledge and domain information.

Example: Agent knowing that Paris is the capital of France without needing to look it up.

Episodic Memory

Specific past experiences and event sequences.

Example: Agent recalling "last time user asked about Python, they preferred beginner-friendly explanations."

Top Frameworks for Building AI Agents



LangChain

Comprehensive framework with extensive tool integration.
Example: Build a chatbot that can search the web, query databases, and call APIs.

LangGraph

Graph-based orchestration with state management and cyclic workflows.
Example: Create approval workflows where documents loop back for revision until approved.

CrewAI

Role-based multi-agent collaboration with task delegation.
Example: Simulate a marketing team with researcher, copywriter, and designer agents working together.

AutoGen

Multi-agent conversations with code execution capabilities.
Example: Build coding assistant where agents debate solutions and execute code to verify correctness.

Agno Agents

Enterprise-ready platform with production focus.
Example: Deploy scalable customer service agents with monitoring, version control, and compliance features.

No-Code Platforms for AI Agents

01

n8n

Visual workflow automation with 400+ integrations and AI nodes. Example: Create workflow that monitors emails, uses AI to categorize them, and auto-responds to common queries.

Make

1500+ app integrations with intuitive visual interface. Example: Build automation that scrapes product data, analyzes it with AI, and updates your inventory system.

02

03

Zapier

6000+ app connections with simple trigger-action model. Example: When form submitted, AI agent enriches lead data and adds to CRM automatically.

Flowise

Open-source LangChain visual builder for custom agents. Example: Drag-and-drop to build a customer support bot with document retrieval and conversation memory.

04